such as WikiData$_{recent}$, ZsRE, and WikiData$_{counterfact}$, we compute the metric as [69] which computes the accuracy of the outputs. Suppose x_k is the expression for the updated knowledge k and y_k^* is the corresponding target output for editing.

$$\text{Edit Succ.} = \sum_{(x_k, y_k^*)} \mathbb{1}\{\text{argmax}_y f_{\theta'}(y \mid x_k) = y_k^*\} \tag{10}$$

Also, for portability, we compute the post-edited model's performance on the given sets. As to the calculation of locality, some work computes the post-edited model's performance on the locality set $O(x_k)$. Here, for a better comparison, we test whether the model keeps its original answer.

$$\text{Locality} = \mathbb{E}_{x_k, y_k^* \sim O(x_k)} \mathbb{1}\{f_{\theta'}(y \mid x_k) = f_\theta(y \mid x_k)\} \tag{11}$$

Meanwhile, for the sentiment edit task Convsent, we compute the Edit Succ. and Locality as the original dataset [153]:

$$\text{Edit Succ.}_{\text{Convsent}} \triangleq \mathbf{z}_{\text{sentiment}} \cdot \mathbf{z}_{\text{topic}} \tag{12}$$

Where $\mathbf{z}_{\text{sentiment}}$ goes to one if the edited model generates correct sentiment responses and $\mathbf{z}_{\text{topic}}$ one if the edited model's answer related to the target topic. The locality of Convsent is computed as the KL-divergence so the lower the number, the better the performance is:

$$\text{Locality}_{\text{Convsent}} \triangleq \mathbb{KL}(f_\theta(\cdot \mid x_k) \| f_{\theta'}(\cdot \mid x_k)) \tag{13}$$

For the knowledge erasure task Sanitation, we calculate edit success as whether the model answers "I don't know." for the given knowledge. As for the locality, we compute the performance on the retain sets as to whether the model keeps their original answer.

4.2 Main Results

We list the results of current knowledge editing methods on **Llama2-7b-chat** in Table 4.

DataSet	Metric	SERAC	ICE	AdaLoRA	MEND	ROME	MEMIT	FT-L	FT-M
WikiData$_{recent}$	Edit Succ. ↑	98.68	60.74	100.00	95.75	97.18	97.05	55.75	100.00
	Portability ↑	63.52	36.93	64.69	55.88	55.25	56.37	40.86	65.44
	Locality ↑	100.00	33.34	56.42	94.76	54.77	52.15	43.70	64.33
	Fluency ↑	553.19	531.01	579.57	557.11	579.66	573.89	529.24	574.32
ZsRE	Edit Succ. ↑	99.67	66.01	100.00	96.74	96.77	95.37	53.93	99.98
	Portability ↑	56.48	63.94	58.03	60.41	52.63	52.67	45.64	60.31
	Locality ↑	30.23	23.14	75.76	92.79	53.67	48.32	73.42	89.78
	Fluency ↑	410.89	541.14	563.56	524.33	573.75	563.31	493.01	552.26
WikiBio	Edit Succ.↑	99.69	95.53	100.00	93.66	96.08	94.40	66.33	100.00
	Locality ↑	69.79	47.90	81.28	69.51	62.74	61.51	79.86	93.38
	Fluency ↑	606.95	632.92	618.45	609.39	617.69	616.65	606.95	612.69
WikiData$_{counterfact}$	Edit Succ. ↑	99.99	69.83	100.00	80.03	98.57	98.05	45.15	100.00
	Portability ↑	76.07	45.32	69.89	52.01	55.92	58.56	33.60	74.36
	Locality ↑	98.96	32.38	70.31	94.38	51.97	46.62	50.48	76.76
	Fluency ↑	549.91	547.22	580.29	555.72	584.04	575.96	528.26	575.62
ConvSent	Edit Succ. ↑	62.75	52.78	44.89	50.76	45.79	44.75	49.50	46.10
	Locality ↓	0.26	49.73	0.18	3.42	0.00	0.00	0.00	0.00
	Fluency ↑	458.21	621.45	606.42	379.43	606.32	602.62	607.86	592.52
Sanitation	Edit Succ. ↑	0.00	72.50	2.50	0.00	85.00	48.75	0.00	75.00
	Locality ↑	100.00	56.58	65.50	5.29	50.31	67.47	14.78	47.07
	Fluency ↑	416.29	794.15	330.44	407.18	465.12	466.10	439.10	416.29

Table 4: Results of existing knowledge editing methods on KnowEdit. We have updated the results after optimizing certain methods (related to AdaLoRA) and fixing computational bugs (related to ROME and MEMIT) in the EasyEdit tool. These improvements have led to better results than before. The symbol ↑ indicates that higher numbers correspond to better performance, while ↓ denotes the opposite, with lower numbers indicating better performance. The locality of Convsent is computed as the KL-divergence so the lower the number, the better the performance is. For WikiBio and Convsent, we do not test the portability as they are about specific topics.

Considering the overall performance across various knowledge editing tasks, our newly proposed FT-M implementation outperforms other methods, highlighting the effectiveness of fine-tuning the

model on specific parameters. However, all current knowledge editing methods suffer from low portability performance, indicating a need for further improvements in this area.

Regarding knowledge editing methods, SERAC demonstrates strong performance for tasks involving knowledge insertion and modification. Its edit success rate is better than other editing methods, and the portability is relatively good as the new counterfact model can learn the edited knowledge effectively. Meanwhile, without changing the original model's parameters, SERAC obtains a good locality performance except for ZsRE. However, since the counterfact model is usually smaller than the original model, its generation ability is not that strong, and here, We can find SERAC's fluency for WikiData$_{counterfact}$, ZsRE, and Convsentis lower than other editing methods like MEND. Meanwhile, for ICE, we can find that the edit success is not that good, which may be attributed to the knowledge conflict problem. Meanwhile, IKE proposed to concatenate demonstrations as the prompt, but they required a long input length and limited the model to conducting downstream tasks.

For the methods that edit the model's parameters, we can find that MEND obtains good performance across these tasks in different metrics. Its edit success and portability are good and demonstrate good locality and fluency. While for ROME and MEMIT, despite the better edit success, their locality is not as good as MEND and other type of editing methods. Meanwhile, its portability is unsatisfactory. For the local fine-tune method FT-L, its edit success is not as good as ROME or MEMIT, however, the locality and portability are better. Also, it seems that FT-M can deal with insertion tasks better as its edit success and portability for WikiData$_{recent}$ is better than ZsRE and WikiData$_{counterfact}$. For the WikiBio task, current methods can alleviate hallucination properly and maintain good fluency. As to the task Convsent, we find that current methods cannot change the model's sentiment well as the edit success is lower than 65%. SERAC, which can deal with small LMs perfectly [153], performs not that well on the 7B model. MEND also shows low fluency for these tasks considering its great performance for fact-level editing in other tasks. As to the knowledge erasure task Sanitation, which aims to erase knowledge from LLMs, we can find that current knowledge editing methods cannot tackle this task properly. We can find that ROME can refrain from the model not providing the target knowledge as it gets 90% accuracy. However, it would destroy the model's performance on unrelated knowledge because its locality is just 55.61%. Other editing methods cannot erase the model related to the given knowledge either.

We also show the average performance of results on WikiData$_{recent}$ and WikiData$_{counterfact}$ in sub-metrics of portability and locality, as we discussed in the previous evaluation part in Figure 3. Here, we can find that MEND performs better under the reasoning set, while AdaLoRA shows good logical generalization performance.

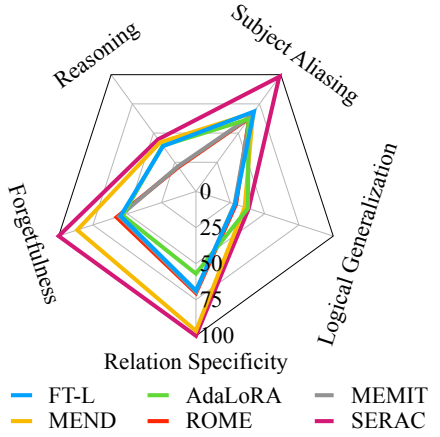


Figure 3: Average sub-metrics performance of results on several fact edit datasets in Portability and Locality.

4.3 Impact on General Tasks

In this Section, we explore the impact of applying knowledge editing methods on the performance of a language model across various domains. Our main goal is to determine if incorporating edits related to specific factual knowledge can unintentionally hinder the model's proficiency in unrelated areas. We select a series of benchmarks that cover areas such as commonsense reasoning, general intelligence, and world knowledge. These benchmarks include CommonsenseQA [199], PIQA [200], Xsum [201], and TriviaQA [202], as well as specific tasks from the MMLU [203] and AGIEval [204] suites, which are known for their distinguished evaluation criteria suites. All evaluations are conducted using the OpenCompass tool [205], ensuring a standardized testing environment. We report the ROUGE-1 here for Xsum. The edited models are evaluated in a zero-shot setting on these tasks after being sequentially modified with five factual updates. An intriguing observation from Table 5 is that, on a holistic level, the edited models man-

	CommonsenseQA	PIQA	TriviaQA	X_Sum	MMLU	AGIEval
Llama2-Chat	49.55	64.91	45.39	22.34	6.87	27.81
FT-L	50.78	67.79	34.60	22.31	7.64	28.56
MEND	49.80	65.23	45.63	22.09	7.64	27.49
ROME	48.89	65.45	45.19	22.46	7.43	27.38
MEMIT	49.80	65.12	45.26	22.34	7.00	28.27
AdaLoRA	49.39	65.07	45.29	22.31	6.90	27.72

Table 5: The zero-shot performance on the general LLM benchmark with Llama2-Chat-7B as the base model. Here, we conduct 5 consecutive edits for each method using the Wiki$_{recent}$ dataset to evaluate the post-edited model's general ability. We adopt the OpenCompass [205] to evaluate the model and use the HuggingFace setting. The MMLU and AGIEval are both the average performance of the sub-tasks.

Method		ZsRE⇒ Wiki$_{recent}$	Wiki$_{recent}$ ⇒ Wiki$_{counterfact}$	Wiki$_{recent}$ ⇒ ZsRE
MEND	Edit Succ.	95.91	66.15	89.79
	Portability	61.80	45.95	54.36
	Locality	66.57	94.83	95.80
	Fluency	554.28	592.82	571.39
SERAC	Edit Succ.	97.42	99.43	99.31
	Portability	60.42	68.85	57.70
	Locality	27.25	100.00	79.04
	Fluency	487.29	552.51	511.95

Table 6: Cross-Domain Editing Results. Performance (accuracy) of the compared methods, which are firstly trained on a source dataset and then directly conduct prediction on a target dataset (denoted as source ⇒ target).

aged to sustain a performance level that is close to their unedited counterparts. This suggests that the negative impact of the editing was limited to directly altered topics. However, one exception to this trend is the FT-L model's performance on TriviaQA, which shows a noticeable decline from an initial score of 45.39 to 34.60 after the edit. Nevertheless, taking a broader perspective, we can observe commendable consistency. This implies that contemporary knowledge editing methods are effective in executing five targeted factual updates with minimal disruptions to the model's cognitive capabilities and adaptability across diverse knowledge domains.

4.4 Multi-Task Knowledge Editing

Previous work considered a sequential edit [163, 161, 69] for a lifelong knowledge editing. However, they always conduct sequential editing on a single dataset from the same distribution. This is a bit different from Continuous learning. Knowledge editing is not a task focusing on single-domain knowledge or fact. In reality, we may want to modify our model from different perspectives from different distributions [206].

Cross-domain Editing Both MEND and SERAC methods rely on a training dataset to help the model learn how to edit parameters. We evaluate their performance in a cross-domain setting and present the results in Table 6.

For the MEND method, the hyper-network trained using the ZsRE dataset exhibits better cross-domain performance than that trained with the recent dataset. This can be attributed to the enormous size of the ZsRE dataset, allowing MEND's hyper-network to enhance its parameter-editing capabilities. Meanwhile, the SERAC approach, by leveraging its cache, exhibits significant cross-domain editing prowess.

Continual Editing Methods like LoRA and ROME do not require a training set and can be applied directly to different domains. Hence, we consider a more challenging setting for continual